

Boolean Functions: SOP, POS, Minterms & Maxterms

Truth table $\leftrightarrow$ expression $\cdot$ minterm & maxterm $\cdot$ Σ and Π notation $\cdot$ canonical forms $\cdot$ don't-cares

QUICK TRICK

Main idea of this chapter: read the **1s** of a truth table to get SOP. Read the **0s** (and flip) to get POS. The tricks are in the green boxes.

6.1 What is a Boolean function?

● LEVEL 1 · BEGINNER

What is it? A Boolean function gives **one output (0 or 1)** for every possible combination of inputs. Example: $F(A, B, C)$ means the output F depends on three inputs A , B and C .

Three ways to show the same function:

Way	Example
Truth table	A list of all input combinations and the output for each
Boolean expression	$F = A \cdot B' + C$
Logic circuit	AND, OR and NOT gates joined together

Why do we need to switch between them? A problem is usually given in words or as a truth table. We must turn it into an expression, simplify it, and then build the circuit.

EASY MEMORY

n inputs $\rightarrow 2^n$ rows in the truth table. 2 inputs = 4 rows, 3 inputs = 8 rows, 4 inputs = 16 rows.

Expression $\rightarrow$ Truth table

Steps: write all input combinations (count in binary: 000, 001, 010 ...), work out each part of the expression, then combine. Example: $F = A \cdot B' + C$

A	B	C	B'	$A \cdot B'$	$F = A \cdot B' + C$
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

QUICK TRICK

Quick method: do not fill every column. Find where each product term is 1. $A \cdot B'$ is 1 only when $A = 1$ and $B = 0$ (rows 100, 101). C is 1 whenever $C = 1$ (rows 001, 011, 101, 111). F is 1 on **all rows where at least one term is 1**: rows 1, 3, 4, 5, 7.

6.2 SOP and POS

● LEVEL 1 · BEGINNER

	SOP — Sum of Products	POS — Product of Sums
Meaning	An OR of AND terms	An AND of OR terms
Example	$AB + A'C + BC'$	$(A + B)(A' + C)(B + C')$
Shape of circuit	AND gates → one OR gate	OR gates → one AND gate
You read it from the truth table	From the rows where $F = 1$	From the rows where $F = 0$

EASY MEMORY

Sum = OR (+). Product = AND (·). “Sum of products” = OR of ANDs. “Product of sums” = AND of ORs.

COMMON MISTAKE

$AB + AC$ is SOP. **$(A + B)(A + C)$** is POS. But **$A + BC$** is also SOP (A is a product with one variable). Do not call it POS just because it has brackets somewhere.

6.3 Minterm and Maxterm

● LEVEL 2 · EXAM

Minterm: an **AND** term that contains **every** variable (plain or with a bar). It equals 1 on **exactly one** row.

Maxterm: an **OR** term that contains **every** variable. It equals 0 on **exactly one** row.

Every row gets a number (the row number in binary). Minterm of row i is written m_i and maxterm M_i .

QUICK TRICK

Minterm rule: 1 → plain letter, 0 → letter with bar. (Row 101 → $A B' C$.)

Maxterm rule: 0 → plain letter, 1 → letter with bar. (Row 101 → $A' + B + C'$.)

Maxterm is the opposite of minterm. $M_i = (m_i)'$.

Row	A B C	Minterm	Maxterm
0	000	$m_0 = A'B'C'$	$M_0 = A + B + C$
1	001	$m_1 = A'B'C$	$M_1 = A + B + C'$
2	010	$m_2 = A'BC'$	$M_2 = A + B' + C$
3	011	$m_3 = A'BC$	$M_3 = A + B' + C'$
4	100	$m_4 = AB'C'$	$M_4 = A' + B + C$
5	101	$m_5 = AB'C$	$M_5 = A' + B + C'$
6	110	$m_6 = ABC'$	$M_6 = A' + B' + C$
7	111	$m_7 = ABC$	$M_7 = A' + B' + C'$

EASY MEMORY

minterm = **m**ake it 1 (AND, one row gives 1). **M**axterm = **M**ake it 0 (OR, one row gives 0). Small m → small (AND). Capital M → big (OR).

6.4 Canonical forms and Σ / Π notation

● LEVEL 2 · EXAM

Canonical SOP = OR of all minterms where $F = 1$. **Canonical POS** = AND of all maxterms where $F = 0$. “Canonical” means every term contains every variable.

Short notation: $\Sigma m(\dots)$ (sigma) lists the minterm numbers where $F = 1$. $\Pi M(\dots)$ (pi) lists the maxterm numbers where $F = 0$.

TRUTH TABLE ---> SOP : take the rows where $F = 1$.
Write one MINTERM for each row. OR them together.
TRUTH TABLE ---> POS : take the rows where $F = 0$.
Write one MAXTERM for each row. AND them together.

Worked example: truth table → SOP and POS

Row	A	B	C	F	Term to write
0	0	0	0	0	$M_0 = (A + B + C)$
1	0	0	1	1	$m_1 = A'B'C$
2	0	1	0	1	$m_2 = A'BC'$
3	0	1	1	0	$M_3 = (A + B' + C')$
4	1	0	0	1	$m_4 = AB'C'$
5	1	0	1	0	$M_5 = (A' + B + C')$
6	1	1	0	0	$M_6 = (A' + B' + C)$
7	1	1	1	1	$m_7 = ABC$

SOP (rows with $F = 1$: 1, 2, 4, 7):

$$F = \text{SIGMA } m(1, 2, 4, 7) = A'B'C + A'BC' + AB'C' + ABC$$

POS (rows with $F = 0$: 0, 3, 5, 6):

$$F = \text{PI } M(0, 3, 5, 6) = (A+B+C) (A+B'+C') (A'+B+C') (A'+B'+C)$$

Both are the same function. (This F is 1 when an **odd** number of inputs are 1: the 3-input XOR.)

QUICK TRICK

Quick check: number of minterms + number of maxterms = 2^n . Here $4 + 4 = 8$. ✓

6.5 Converting between forms

● LEVEL 2 • EXAM

QUICK TRICK

$\Sigma \leftrightarrow \Pi$ (the “rest” trick): $\Sigma m(\text{list}) = \Pi M(\text{all the other row numbers})$. Write all row numbers 0 to $2^n - 1$. Cross out the ones in the list. The rest is the other list.

Example (3 variables): $\Sigma m(1, 2, 4, 7) = \Pi M(0, 3, 5, 6)$.

QUICK TRICK

Complement of a function: $F' = \Sigma m(\text{the rows where } F = 0)$. So if $F = \Sigma m(0, 1, 2, 5)$, then $F' = \Sigma m(3, 4, 6, 7)$. In Π form: $F = \Pi M(3, 4, 6, 7)$ and $F' = \Pi M(0, 1, 2, 5)$. **Complement = swap the lists.**

Normal expression $\rightarrow$ canonical form (add the missing variables)

If a term is missing a variable X , multiply it by **$(X + X')$** (this equals 1, so nothing changes). Then expand.

Example 1 (SOP): $F(A,B,C) = A + B'C$

A is missing B and C: $A(B + B')(C + C') = ABC + ABC' + AB'C + AB'C' \rightarrow$ minterms 7, 6, 5, 4

$B'C$ is missing A: $(A + A')B'C = AB'C + A'B'C \rightarrow$ minterms 5, 1

Join all (5 appears twice, keep once): **$F = \Sigma m(1, 4, 5, 6, 7)$**

Check: $A = 1$ gives rows 4-7 ✓. $A = 0$ needs $B'C = 1$, which is row 001 = 1 ✓.

QUICK TRICK

Shortcut with the number of rows: a term with k variables (out of n) covers 2^{n-k} minterms. In 3 variables: A (1 variable) covers $2^2 = 4$ rows. $B'C$ (2 variables) covers $2^1 = 2$ rows. Just list the rows where the term is true.

Example 2 (POS): $F = (A + B)(A' + C)$

$(A + B)$ is missing C: $A + B + CC' = (A + B + C)(A + B + C') \rightarrow$ maxterms M0, M1

$(A' + C)$ is missing B: $A' + C + BB' = (A' + B + C)(A' + B' + C) \rightarrow$ maxterms M4, M6

$F = \Pi M(0, 1, 4, 6)$

Check: $F = 0$ when $A = B = 0$ (rows 0, 1) or when $A = 1$ and $C = 0$ (rows 4, 6). ✓

Example 3 (Level 3): the majority function

$F = 1$ when at least two of A, B, C are 1. Rows: 011, 101, 110, 111 $\rightarrow$ **$F = \Sigma m(3, 5, 6, 7)$**

$F = A'BC + AB'C + ABC' + ABC$. Simplify: $ABC + A'BC = BC$, $ABC + AB'C = AC$, $ABC + ABC' = AB$.

(ABC can be used again and again because $X + X = X$.) **$F = AB + BC + AC$.**

6.6 Don't-care conditions

● LEVEL 2 • EXAM

What is it? Sometimes an input combination **never happens**, or we do not care about the output for it. We mark that output with **X** (or d). **Why useful?** We may choose X to be 0 or 1, whichever makes the expression **simpler**.

Notation: $F = \Sigma m(1, 3, 7) + d(5)$ means F is 1 for rows 1, 3, 7, and we do not care about row 5.

Example: BCD even-digit detector

A BCD digit uses 4 bits (A B C D), so only 0 to 9 appear. Rows 10 to 15 never occur.

$F = 1$ for even digits: $F = \Sigma m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$

Choose $X = 1$ for rows 10, 12, 14 (they are even). Now every even row is 1 and every odd digit (1, 3, 5, 7, 9) is 0.

The simple answer is $F = D'$ (the last bit is 0 for even numbers). One NOT gate! Without don't-cares it would need many gates.

EXAM TRAP

Don't-cares are **not** written in the SOP or POS list as real minterms. Keep them in a separate $d(\dots)$ list. In the final circuit each X becomes either 0 or 1 (depending on how you used it).

6.7 Counting facts that GATE likes

● LEVEL 3 • GATE

Fact	Value
Number of minterms for n variables	2^n
Number of different functions of n variables	2^{2^n} ($n = 3 \rightarrow 256$, $n = 4 \rightarrow 65536$)
Sum of ALL minterms / product of ALL maxterms	$1 / 0$
If F has m minterms, how many does F' have?	$2^n - m$
A product term with k literals in n variables	covers 2^{n-k} minterms

QUICK TRICK

Minterms inside a term: in 4 variables, the term $AB'D'$ has 3 literals, so it covers $2^{4-3} = 2$ minterms ($A = 1$, $B = 0$, $D = 0$, and C can be 0 or 1): m_8 (1000) and m_{10} (1010).

Quick-Solve Tricks — Summary

Situation	Trick
Make SOP from a truth table	Take the rows where $F = 1$. Minterm: 1 $\rightarrow$ plain, 0 $\rightarrow$ bar. OR them.
Make POS from a truth table	Take the rows where $F = 0$. Maxterm: 0 $\rightarrow$ plain, 1 $\rightarrow$ bar. AND them.
Switch $\Sigma \leftrightarrow \Pi$	Use “the rest” of the row numbers.
Complement of F	Swap the Σ and Π lists (zeros become ones).
Missing variable in a term	Multiply by $(X + X')$ and expand. Or list the rows where the term is true.
Rows covered by a term	2^{n-k} rows for k literals.
Don't care	Choose 0 or 1 to make a bigger/simpler group.

GATE QUICK CHECK — Chapter 6

Minterm m_i : AND of all variables. 1 $\rightarrow$ plain, 0 $\rightarrow$ bar. Equals 1 on ONE row.
Maxterm M_i : OR of all variables. 0 $\rightarrow$ plain, 1 $\rightarrow$ bar. Equals 0 on ONE row.
 $M_i = (m_i)'$ $\Sigma m(\text{list}) = \Pi M(\text{rest})$ F' : swap the lists
Rows = 2^n Functions = $2^{(2^n)}$ k literals cover $2^{(n-k)}$ rows

Traps: using the wrong bar rule for maxterms · reading the 1s when asked for POS · forgetting the missing variable · putting don't-cares in the Σ list.

Practice questions

● LEVEL 1 · BEGINNER

Q1. How many rows are there in the truth table of a 4-variable function?

Q2. Write the minterm m_5 for variables A, B, C.

Q3. Write the maxterm M_3 for variables A, B, C.

● LEVEL 2 · EXAM

Q4. Convert $F(A,B,C) = \Sigma m(1, 2, 5)$ to Π form.

Q5. Write $F = AB + C'$ as Σm (three variables A, B, C).

Q6. A function is 1 only on rows 3, 5, 6, 7 (three variables). Write the canonical SOP and the simplified SOP.

● LEVEL 3 · GATE

Q7. In 4 variables (A, B, C, D), which minterms does the product term $AB'D'$ contain?

Q8. $F(A,B,C) = \Sigma m(0, 1, 2, 5)$. Write F' as Σm and F as ΠM .

Q9. $F(A,B,C) = \Pi M(0, 3, 5, 6)$. Write F as Σm .

Q10. How many different Boolean functions exist for 4 variables?

Q11. $F = \Sigma m(1, 3, 7) + d(5)$ for A, B, C. Find the simplest expression.

Q12. $F(A,B,C,D) = \Sigma m(0, 2, 4, 6, 8, 10, 12, 14)$. Simplify.

Answers

Q	Solution
1	$2^4 = \mathbf{16}$ rows.
2	$5 = 101 \rightarrow 1$ plain, 0 bar, 1 plain $\rightarrow \mathbf{AB'C}$.
3	$3 = 011 \rightarrow 0$ plain, 1 bar, 1 bar $\rightarrow \mathbf{A + B' + C'}$.
4	All rows 0-7. Remove 1, 2, 5 $\rightarrow \mathbf{\Pi M(0, 3, 4, 6, 7)}$.
5	AB is true on rows 110, 111 $\rightarrow 6, 7$. C' is true on rows 000, 010, 100, 110 $\rightarrow 0, 2, 4, 6$. Join $\rightarrow \mathbf{\Sigma m(0, 2, 4, 6, 7)}$.
6	Canonical: $\mathbf{A'BC + AB'C + ABC' + ABC}$. Simplified (majority): $\mathbf{AB + BC + AC}$.
7	$A = 1, B = 0, D = 0, C$ free $\rightarrow 1000$ and $1010 \rightarrow \mathbf{m8 \text{ and } m10}$ (2 minterms; $2^{4-3} = 2$).
8	F' has the other rows: $\mathbf{F' = \Sigma m(3, 4, 6, 7)}$. $F = \mathbf{\Pi M(3, 4, 6, 7)}$.
9	Rest of {0-7} without 0, 3, 5, 6 is 1, 2, 4, 7 $\rightarrow \mathbf{\Sigma m(1, 2, 4, 7)}$.
10	$2^{2^4} = 2^{16} = \mathbf{65536}$.
11	Take X (row 5) as 1: rows 1, 3, 5, 7 all have $C = 1 \rightarrow \mathbf{F = C}$.
12	All even rows have $D = 0$ and there are 8 of them (all rows with $D = 0$) $\rightarrow \mathbf{F = D'}$.

Next: Chapter 7 — Karnaugh Map (K-map): grouping, prime implicants, SOP and POS minimisation, don't-cares, with quick tricks.